College of Engineering and Management, Kolaghat

Sign2Speech: Real-Time Indian Sign Language to Speech Translation System

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Defining the Core Problem:
A Crisis of Communication & Accessibility

1. The Silent Majority: Demographics & Scarcity

63 Million+

Deaf, Hard-of-Hearing, & Speech-Impaired in India (WHO Est.)

Severe Interpreter Crunch: <300 Certified ISL Interpreters Nationwide. Ratio ~1 : 200,000.

2. The Resource Divide: ASL vs. ISL

Global/ASL Resources (~98% of Digital Tools) ≠ **Indian Sign Language (ISL) Resources** (<2%)

Critical Gap: ISL is linguistically distinct from ASL. Global tools are ineffective in India.

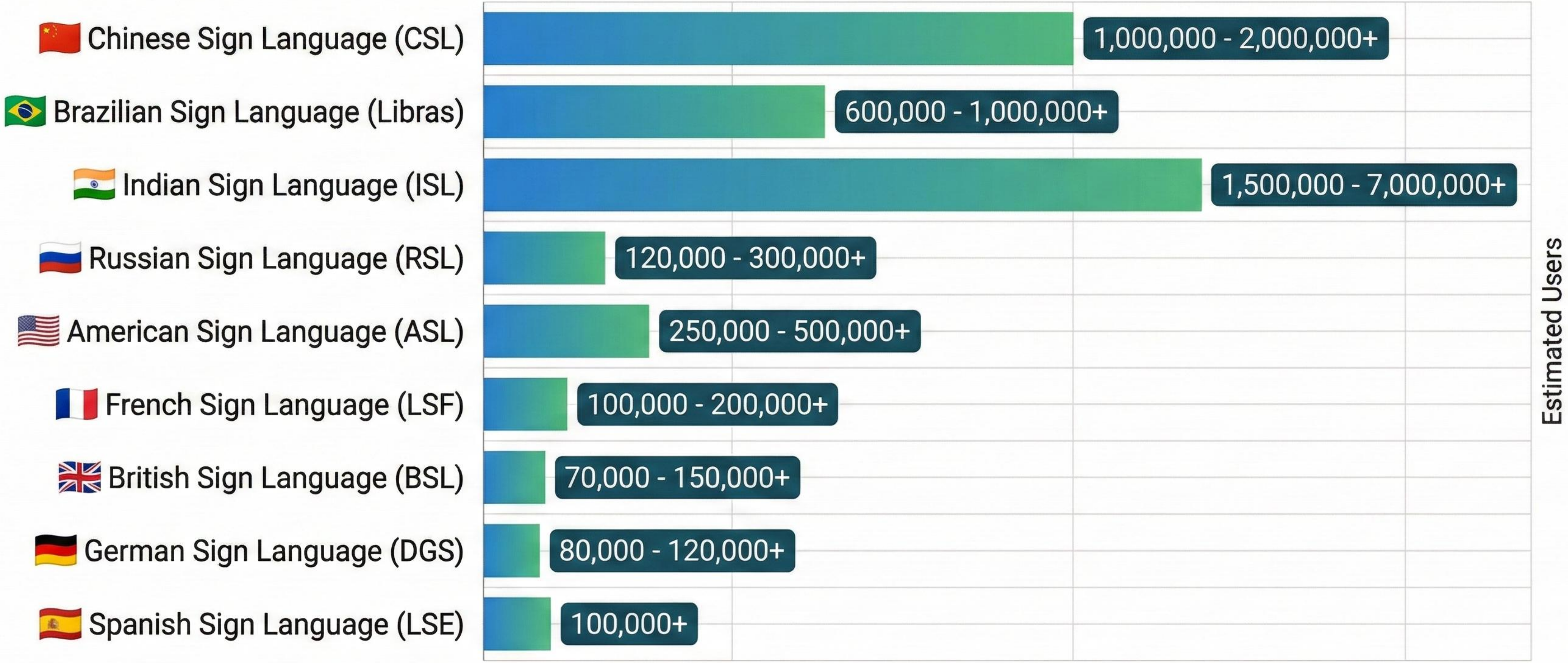
3. The Human Cost: Systemic Exclusion

- **Education Denied**
- **Healthcare Inaccessible**
- **Employment Barriers**
- **Social Isolation**

Consequence: Daily denial of basic rights, information, and opportunity for millions.

Conclusion: The lack of accessible, scalable ISL technology is a fundamental human rights crisis, not just a technical gap.

Global Reach: Number of Users of Major Sign Languages (Estimates)



***Note:** Figures are approximate estimates and vary significantly by source due to challenges in census data collection and differing definitions of 'user'. These numbers often include both deaf and hearing users.

Source: Various academic and organizational estimates.

The Problem: Market Gap & Technological Void

Feature	Google Translate	Existing ISL Apps	Smart Gloves	PROJECT
Input Method	Text/Voice	Text Search	Sensors/Wires	Camera (Vision)
Real-Time AI	✅	❌ (Pre-recorded)	✅	✅
Cost to User	Free	Free	$500+	Free
Supports ISL	❌	✅	❌	✅

Conclusion: No other solution combines real-time AI, zero cost, and ISL support.

Project Objectives

- **Establish a Standardized Data Protocol:** To design and validate a strict data collection pipeline (lighting, distance, and frame filtering rules) to generate high-quality ISL datasets.
- **Develop a Geometry-Based Architecture:** To engineer a lightweight AI model using MediaPipe and Neural Networks that processes skeletal coordinates instead of heavy images.
- **Implement Robust Feature Engineering:** To develop the "Wrist-Scaled Normalization" algorithm to ensure the system works correctly regardless of the user's hand size or distance from the camera.
- **Prototype the Mobile Framework:** To build the core structure of an Offline Android App capable of real-time inference and Text-to-Speech (TTS) integration.

Project Scope

- **Current Scope :**

1. Recognition of Static Hand Gestures (Alphabets & Fixed Signs).
2. Real-time processing on mobile CPUs without internet.
3. Implementation of the Data Collection Protocol.

- **Future Scope :**

1. Integration of Dynamic Gestures (Movement-based signs like "Walking").
2. Expansion of vocabulary to full sentences.
3. Two-way communication (Speech-to-Text).

Technology Stack

1. Machine Learning & Vision (The Core)

- **Language:** Python 3.9
- **Frameworks:** TensorFlow & Keras (Model Training), Google MediaPipe (Skeletal Feature Extraction)
- **Data Processing:** NumPy & Pandas (Vectorization & CSV Handling)

2. Mobile Application (The Deployment)

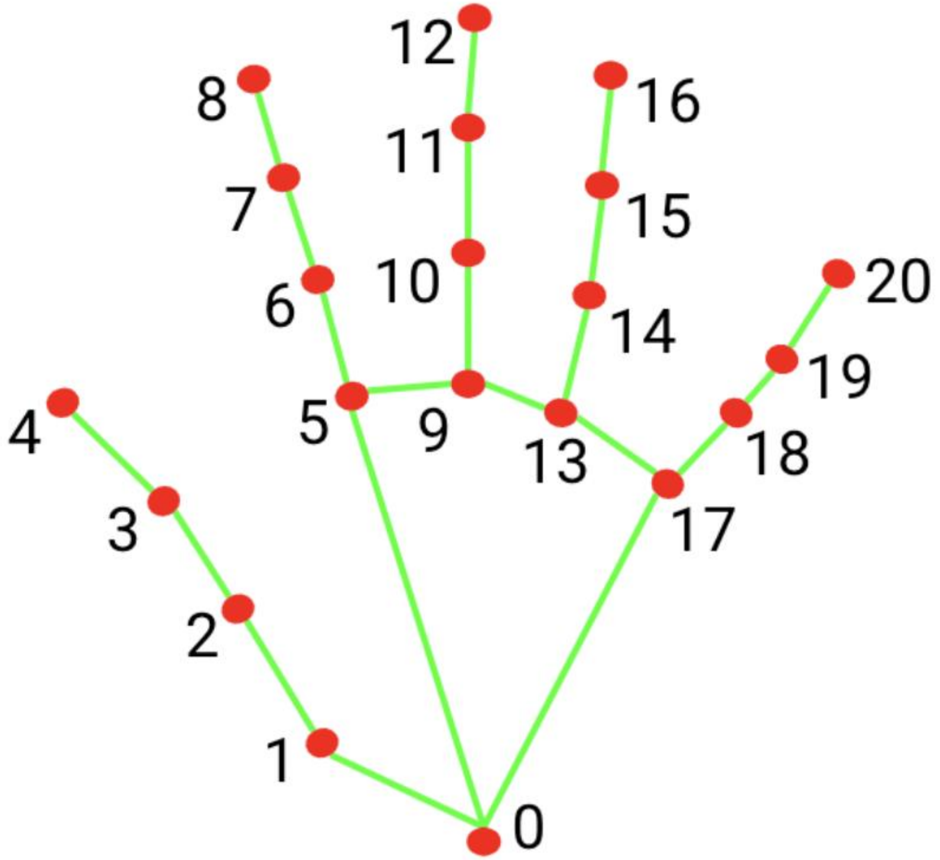
- **Platform:** Android (Native)
- **IDE:** Android Studio
- **Language:** Kotlin (for UI & Logic) & Java
- **Inference Engine:** TensorFlow Lite (TFLite) for offline, on-device prediction
- **Output:** Android Text-to-Speech (TTS) API

3. Development Environment

- **Training:** Google Colab (Cloud-based Model Training)
- **Version Control:** GitHub (Code Management)

4. Hardware Requirements

- **Development:** Standard Laptop with Webcam (No dedicated GPU required)
- **User Device:** Any Android Smartphone (Android 8.0+) with a camera



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

- **MediaPipe Hand Landmark Model** developed by Google. It maps the human hand using 21 distinct 3D coordinates.
- **License:** It is released under the Apache 2.0 License, which is a permissive free software license.
- **Developer:** It was developed and open-sourced by Google in 2019.
- **Codebase:** The full source code is hosted and maintained on GitHub, allowing developers to modify and extend it freely.

Data Structure: The “Skeletal” CSV File

label	right_x_0	right_y_0	right_z_0	...	right_x_20	right_y_20	right_z_20	left_x_0	left_y_0	left_z_0	...	left_x_20	left_y_20	left_z_20
Hello	0.432	0.765	-0.021,	...	0.321	0.654	0.987	0.543	0.678,	-0.012,	...	0.123	0.456	0.789

Target Class
(Sign Name)

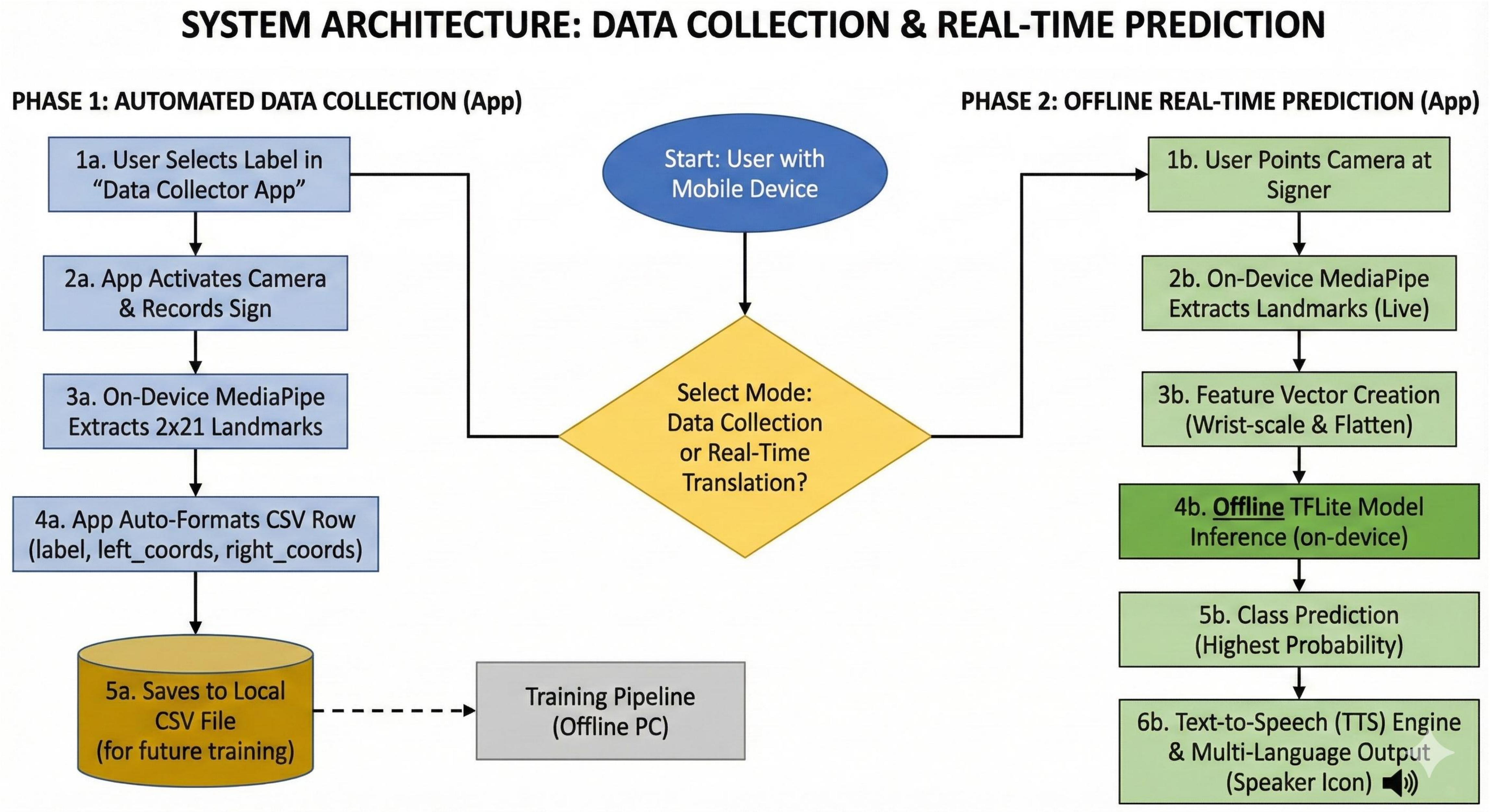
63 Right Hand Features
(21 x 3 coords)

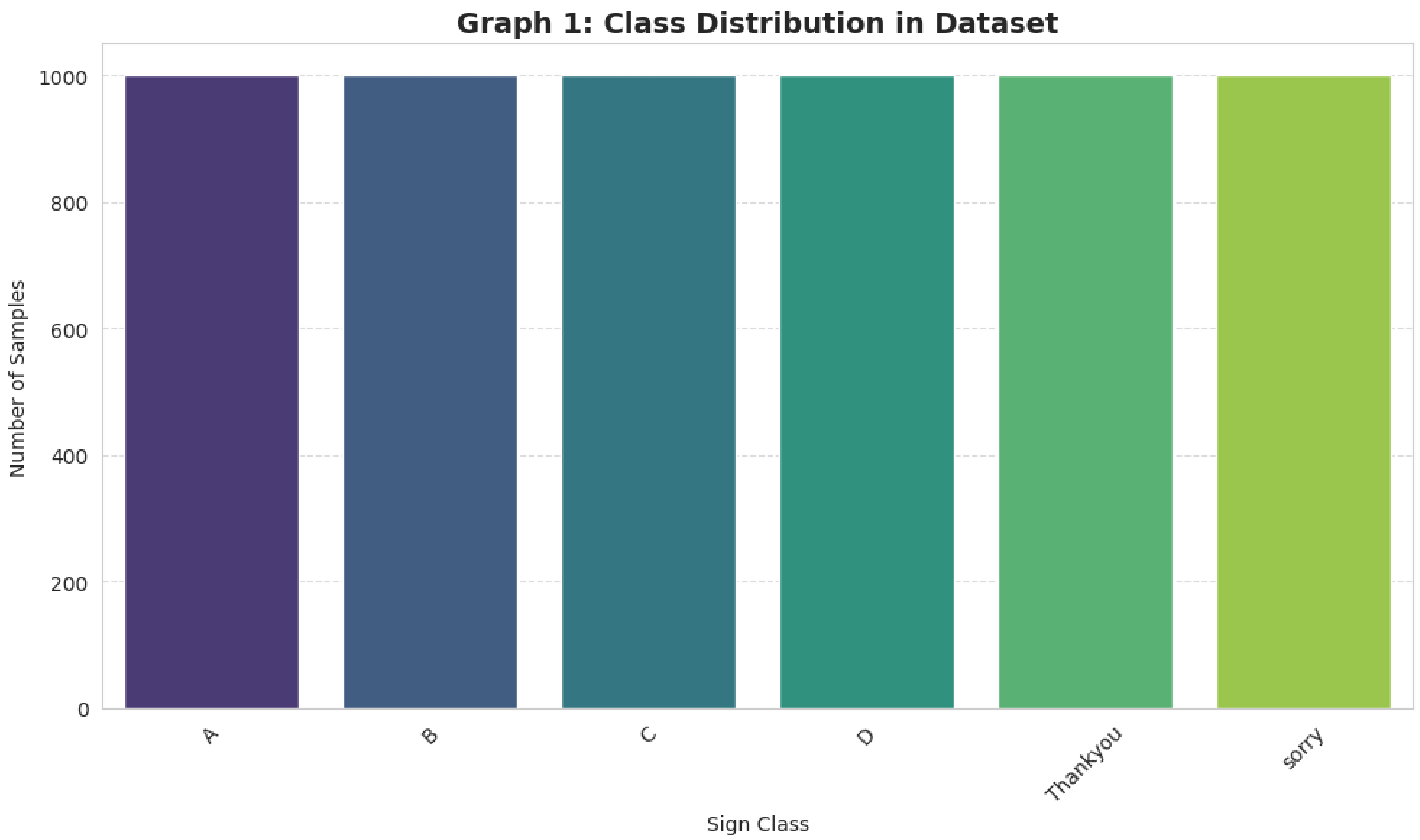
63 Left Hand Features
(21 x 3 coords)

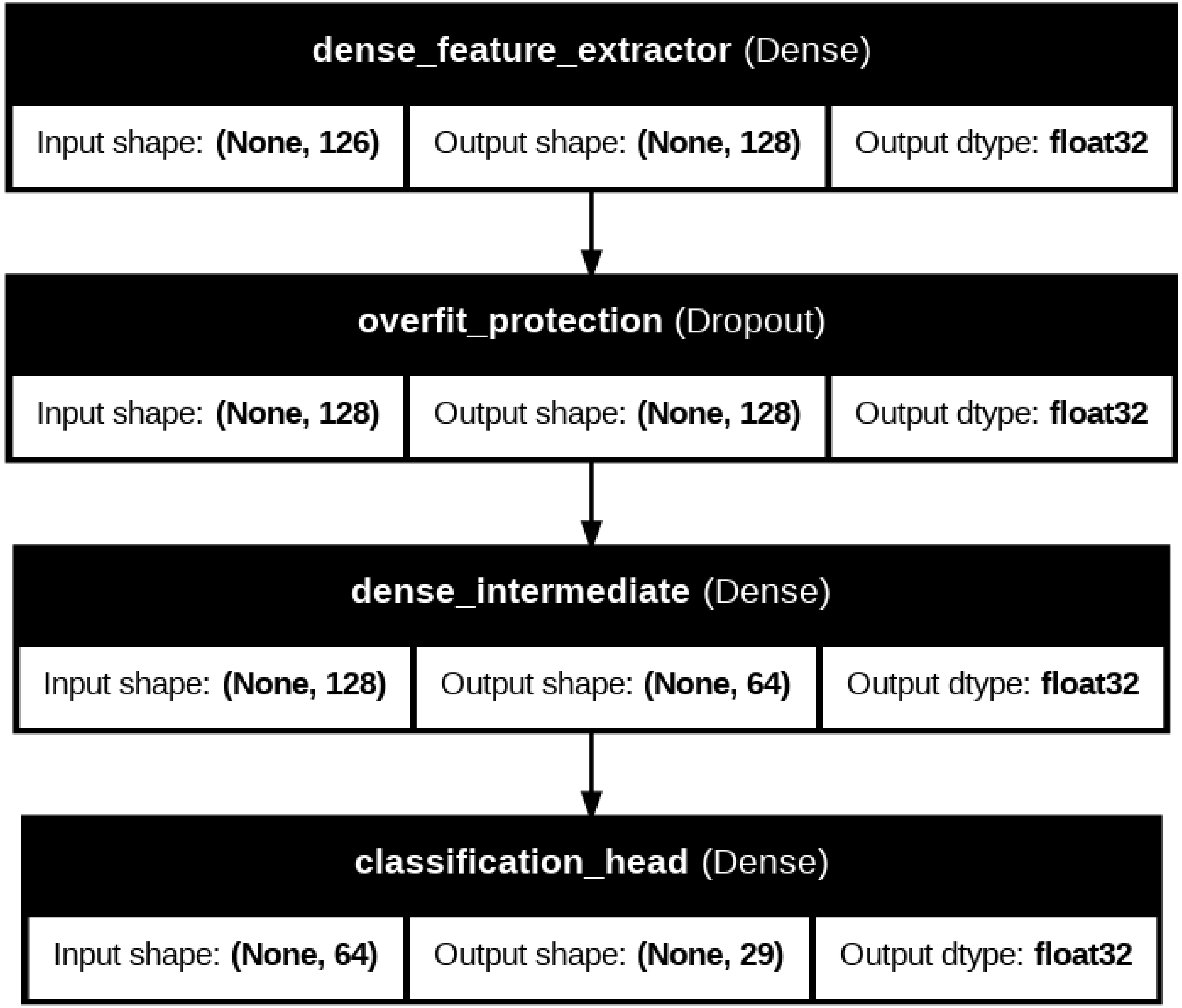
Total: 126 Coordinate Features per Frame (+1 Label)

B

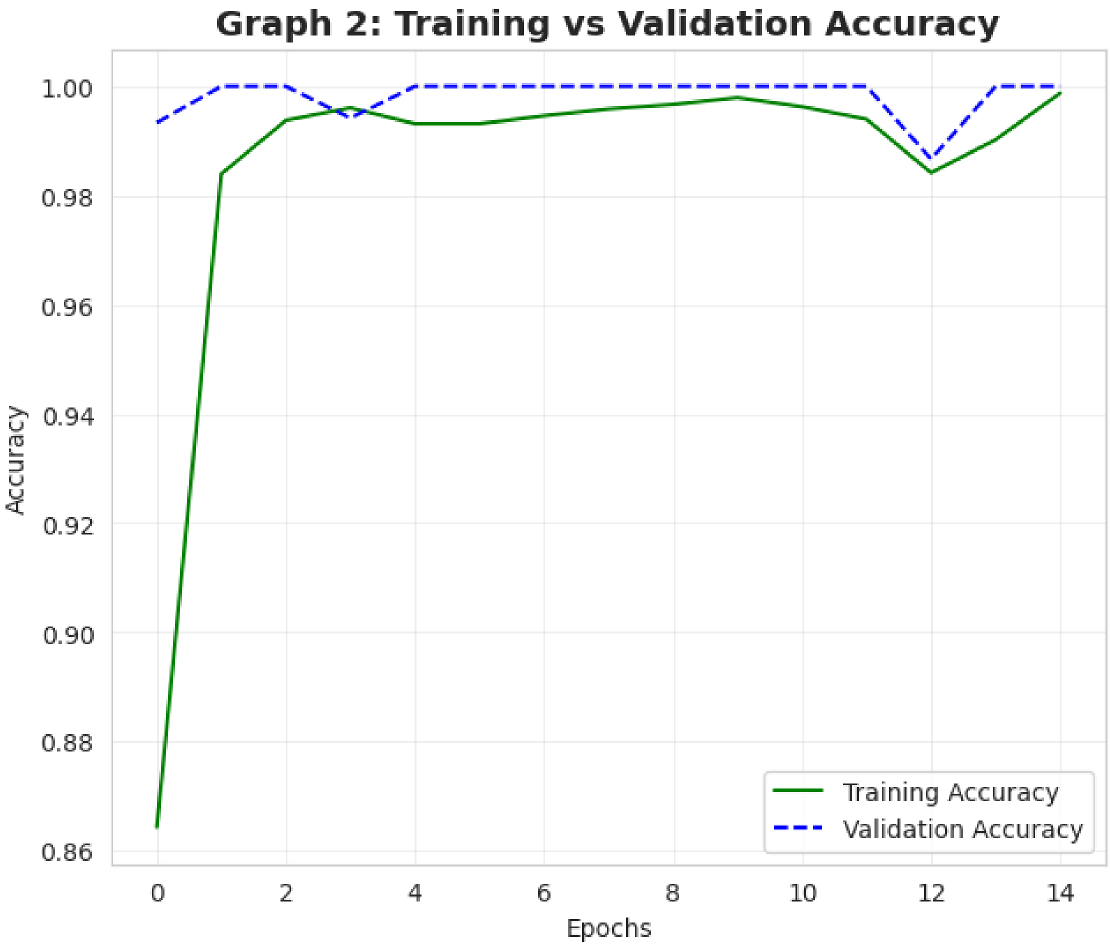
Sorry

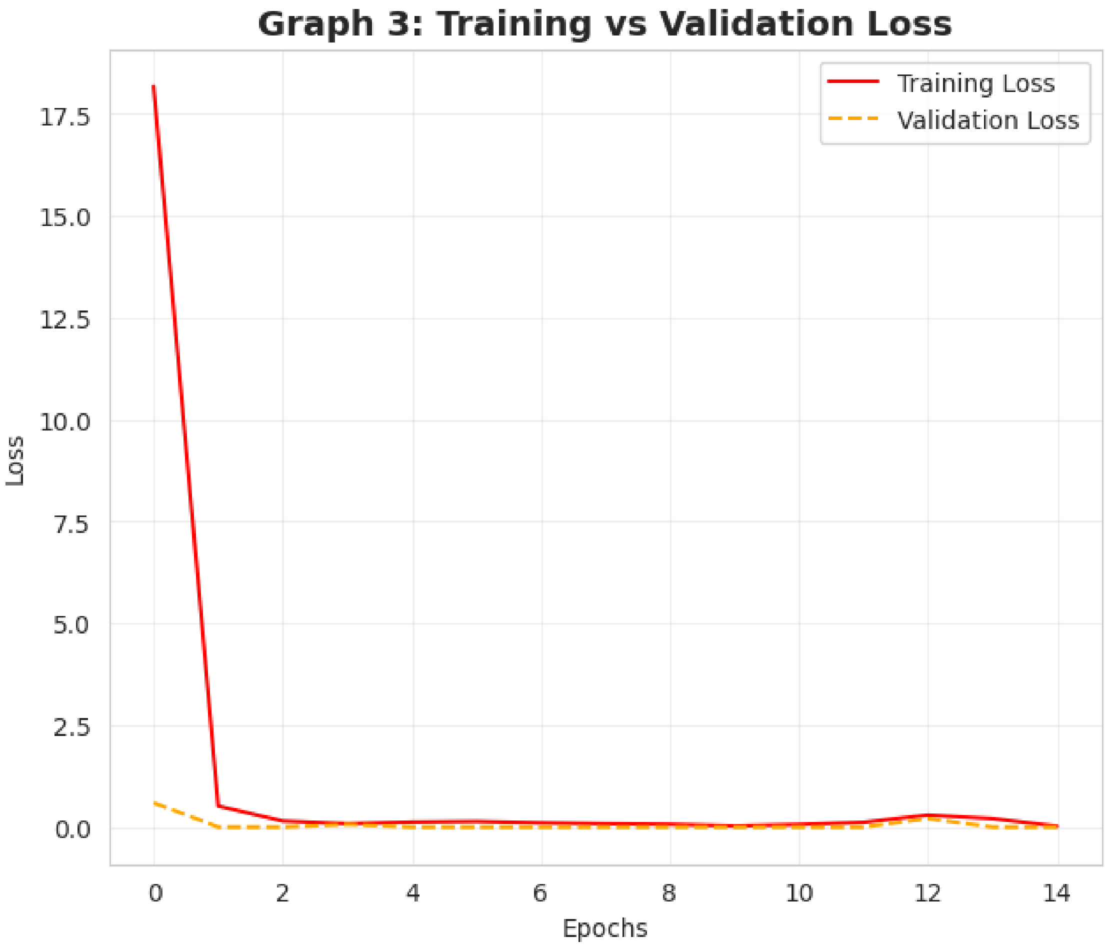

SYSTEM ARCHITECTURE: DATA COLLECTION & REAL-TIME PREDICTION
PHASE 1: AUTOMATED DATA COLLECTION (App)
PHASE 2: OFFLINE REAL-TIME PREDICTION (App)
Start: User with Mobile Device
Select Mode: Data Collection or Real-Time Translation?
1a. User Selects Label in "Data Collector App"
2a. App Activates Camera & Records Sign
3a. On-Device MediaPipe Extracts 2x21 Landmarks
4a. App Auto-Formats CSV Row (label, left_coords, right_coords)
5a. Saves to Local CSV File (for future training)
Training Pipeline (Offline PC)
1b. User Points Camera at Signer
2b. On-Device MediaPipe Extracts Landmarks (Live)
3b. Feature Vector Creation (Wrist-scale & Flatten)
4b. Offline TFLite Model Inference (on-device)
5b. Class Prediction (Highest Probability)
6b. Text-to-Speech (TTS) Engine & Multi-Language Output (Speaker Icon)

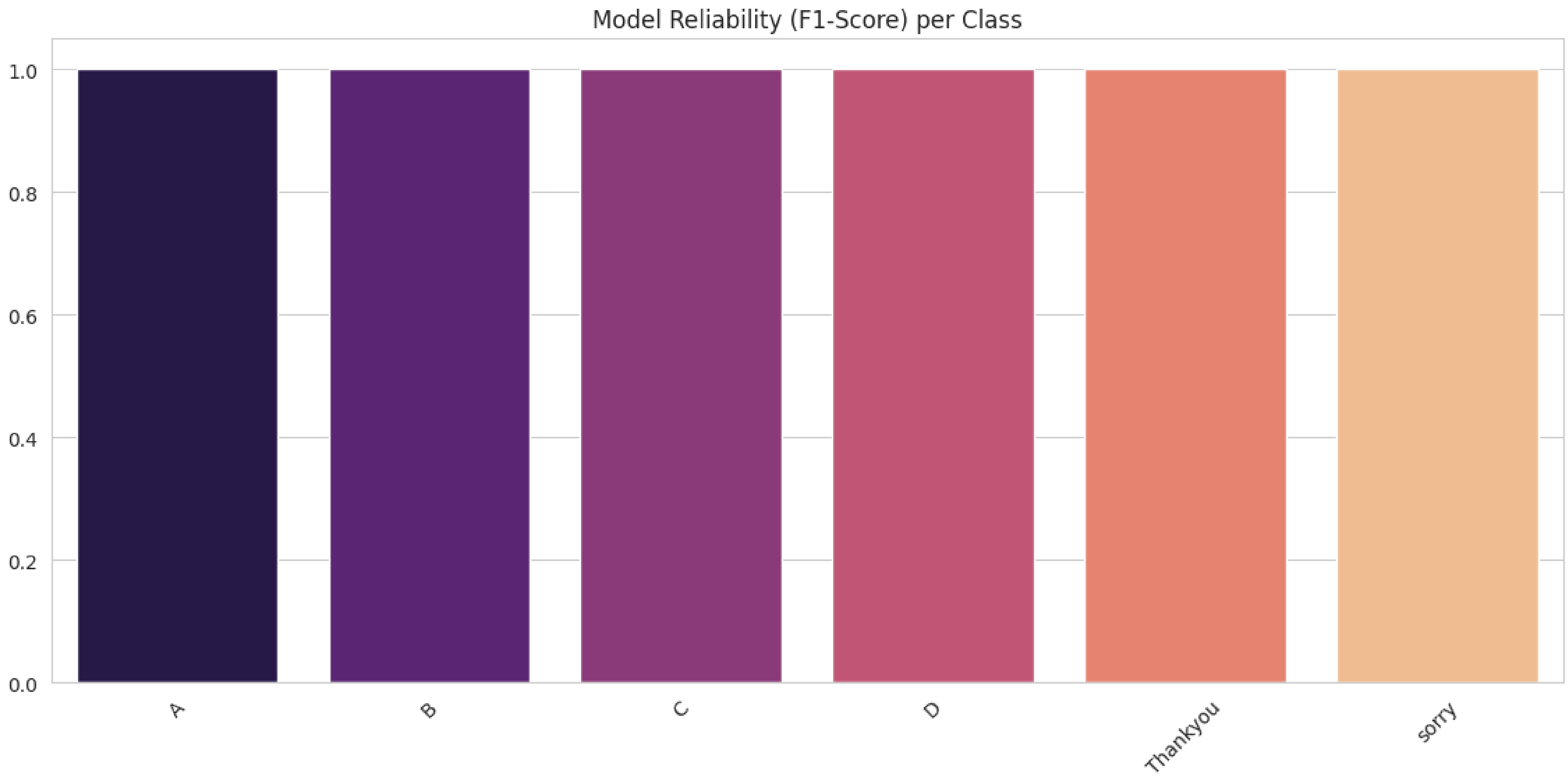

Graph 1: Class Distribution in Dataset
1000
800
600
400
200
0
Number of Samples
A
B
C
D
Thankyou
sorry
Sign Class



dense_feature_extractor (Dense)
Input shape: (None, 126)
Output shape: (None, 128)
Output dtype: float32
overfit_protection (Dropout)
Input shape: (None, 128)
Output shape: (None, 128)
Output dtype: float32
dense_intermediate (Dense)
Input shape: (None, 128)
Output shape: (None, 64)
Output dtype: float32
classification_head (Dense)
Input shape: (None, 64)
Output shape: (None, 29)
Output dtype: float32

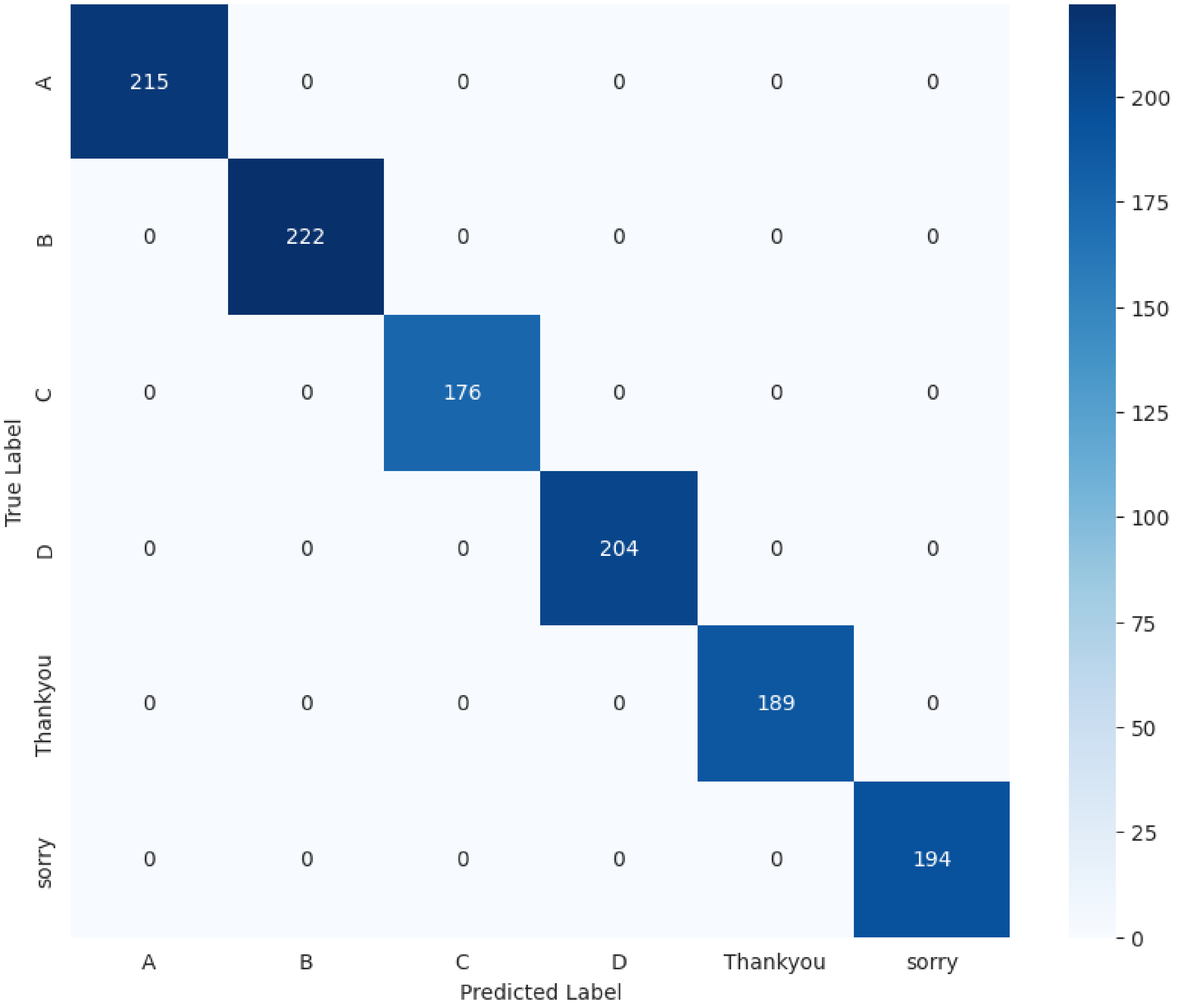

Graph 2: Training vs Validation Accuracy
Accuracy
Epochs
Training Accuracy
Validation Accuracy

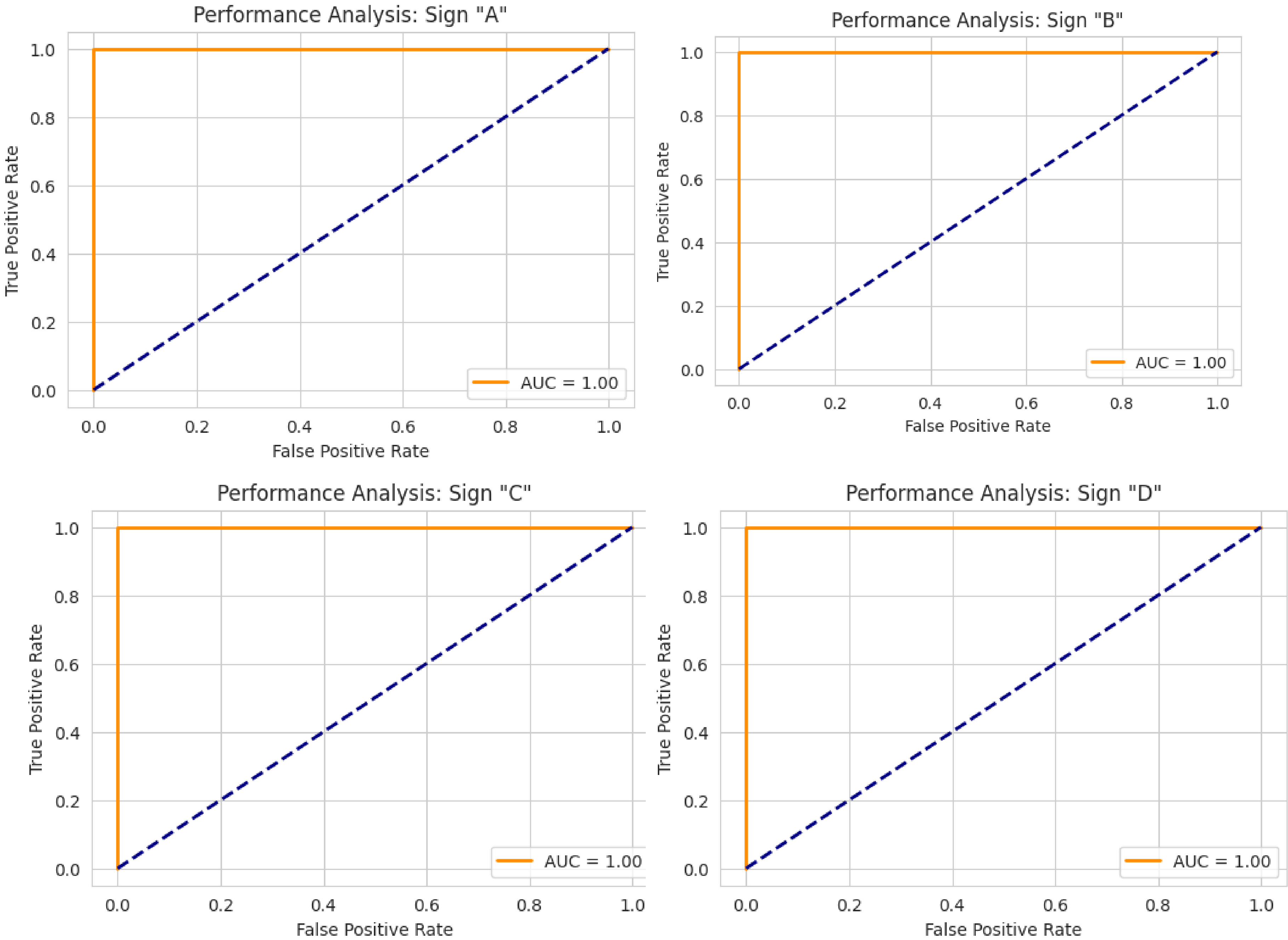

Graph 3: Training vs Validation Loss
Loss
Epochs
Training Loss
Validation Loss

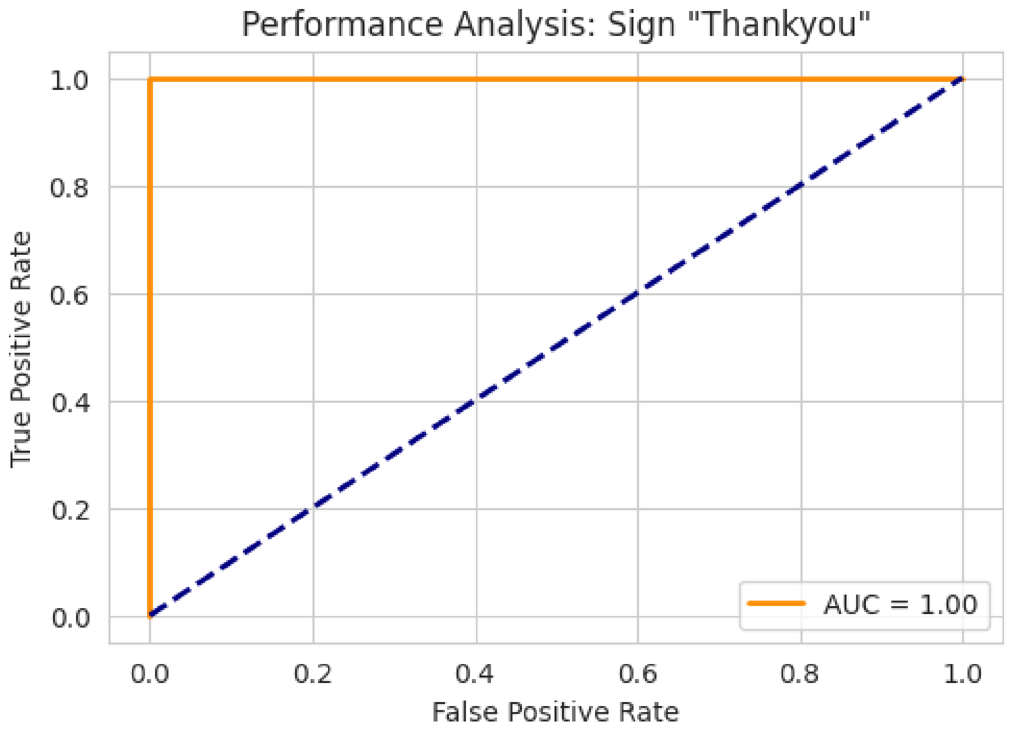


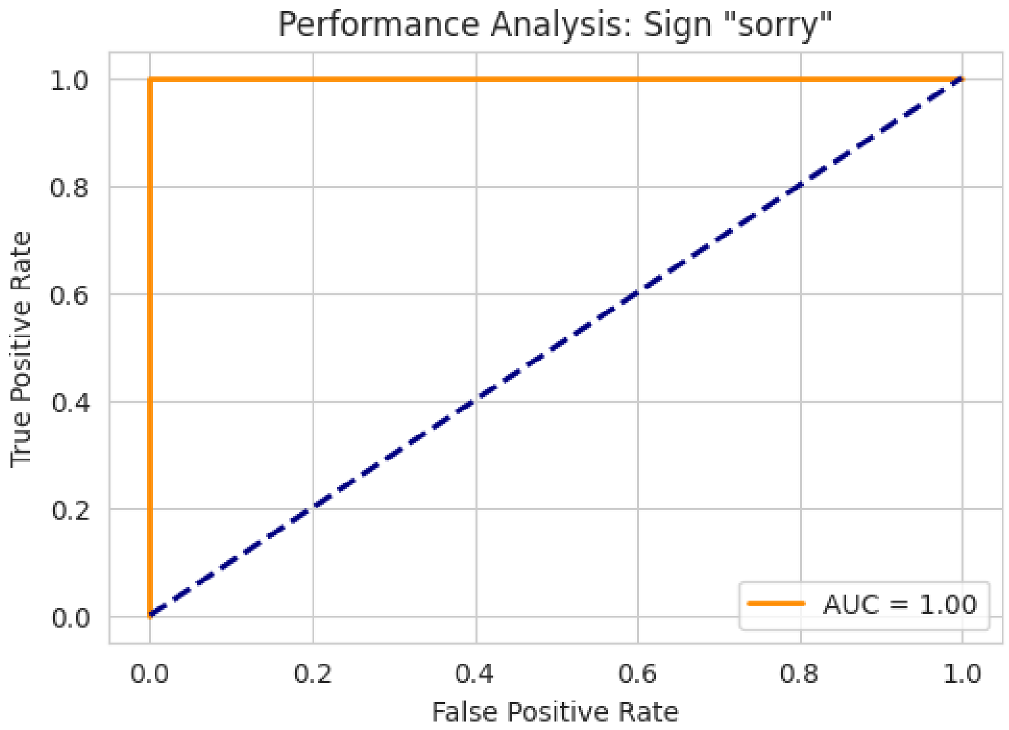
Model Reliability (F1-Score) per Class
1.0
0.8
0.6
0.4
0.2
0.0
A
B
C
D
Thankyou
sorry

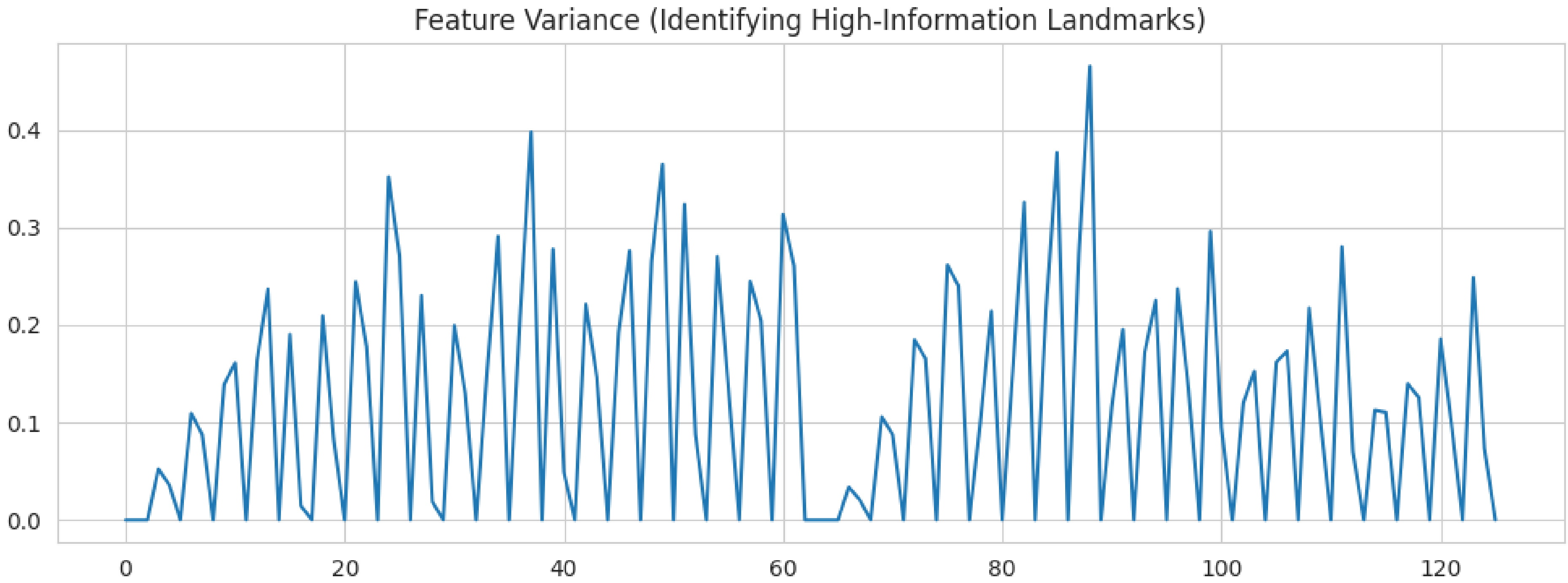
Graph 4: Confusion Matrix



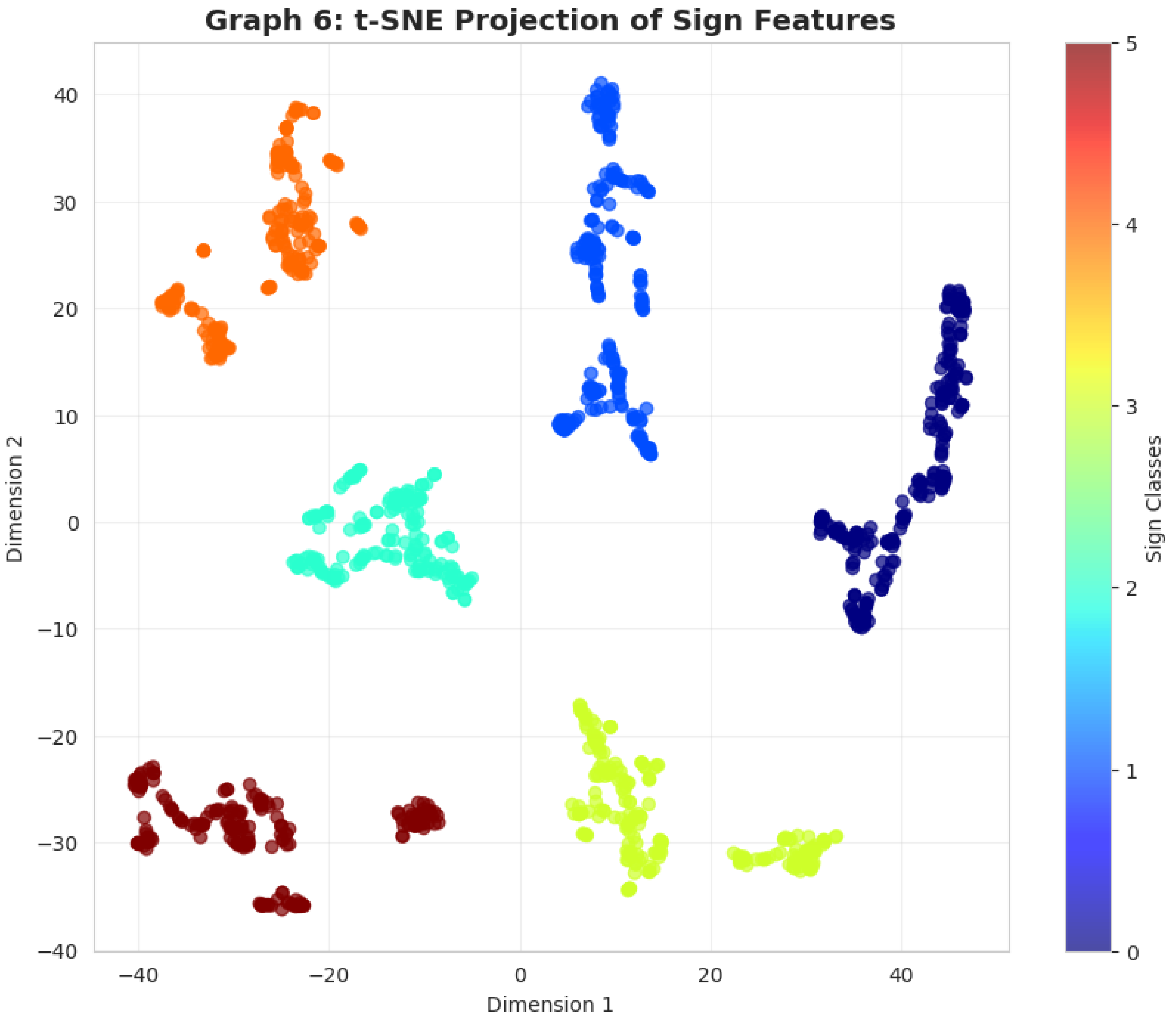

Performance Analysis: Sign "A"
True Positive Rate
False Positive Rate
AUC = 1.00
Performance Analysis: Sign "B"
True Positive Rate
False Positive Rate
AUC = 1.00
Performance Analysis: Sign "C"
True Positive Rate
False Positive Rate
AUC = 1.00
Performance Analysis: Sign "D"
True Positive Rate
False Positive Rate
AUC = 1.00


Performance Analysis: Sign "Thankyou"
True Positive Rate
False Positive Rate
AUC = 1.00
0.0
0.2
0.4
0.6
0.8
1.0


Performance Analysis: Sign "sorry"
True Positive Rate
False Positive Rate
AUC = 1.00
0.0
0.2
0.4
0.6
0.8
1.0



Feature Variance (Identifying High-Information Landmarks)
0.4
0.3
0.2
0.1
0.0
0
20
40
60
80
100
120


Graph 6: t-SNE Projection of Sign Features
Dimension 1
Dimension 2
Sign Classes
−40
−20
0
20
40
−40
−30
−20
−10
0
10
20
30
40
0
1
2
3
4
5

Prototype Demonstration


A

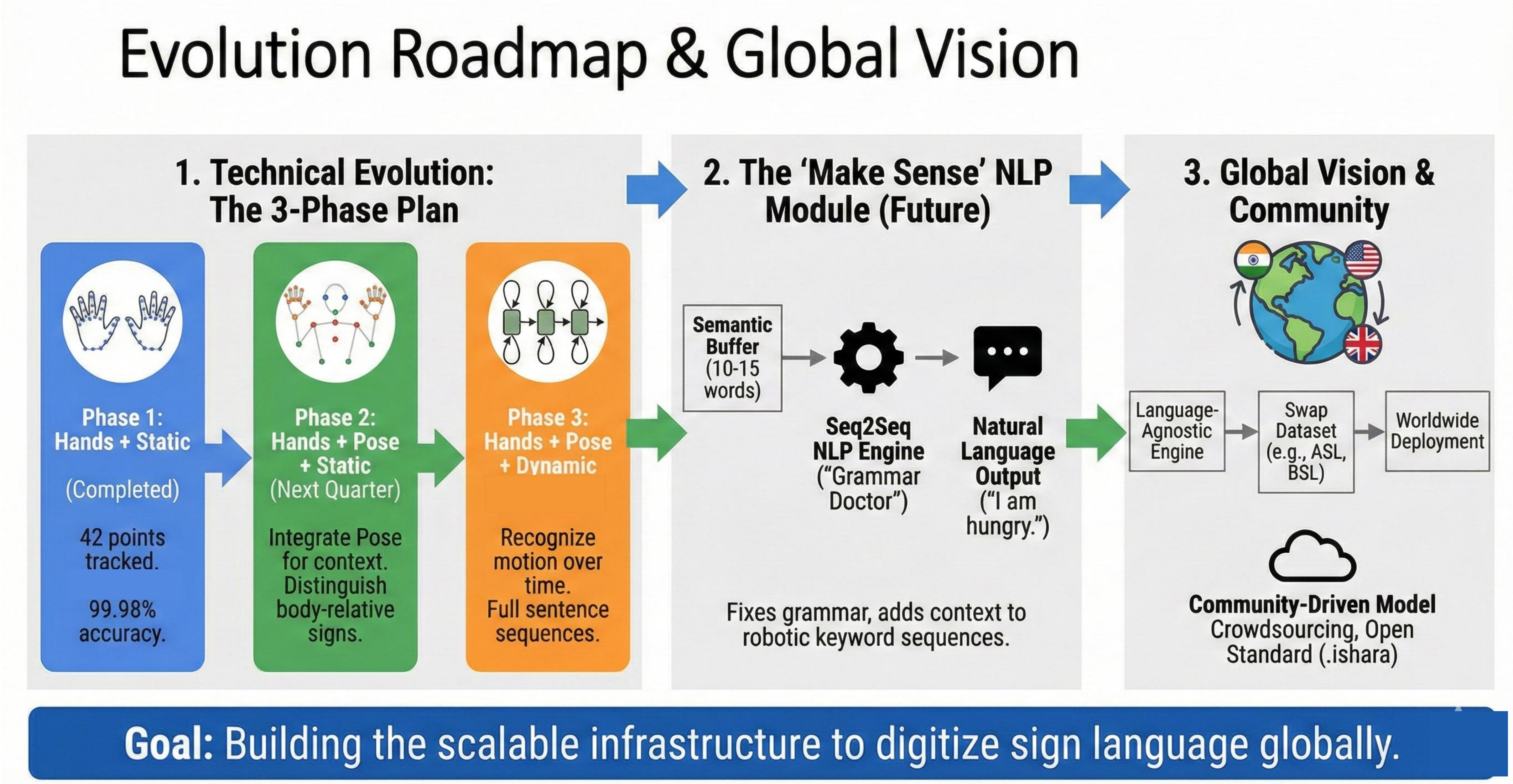
B


DEF
C


D

Sorry


धन्यवाद - Thank You


Evolution Roadmap & Global Vision
1. Technical Evolution: The 3-Phase Plan
Phase 1: Hands + Static
(Completed)
42 points tracked.
99.98% accuracy.
Phase 2: Hands + Pose + Static
(Next Quarter)
Integrate Pose for context. Distinguish body-relative signs.
Phase 3: Hands + Pose + Dynamic
Recognize motion over time. Full sentence sequences.
2. The 'Make Sense' NLP Module (Future)
Semantic Buffer (10-15 words)
Seq2Seq NLP Engine ("Grammar Doctor")
Natural Language Output ("I am hungry.")
Fixes grammar, adds context to robotic keyword sequences.
3. Global Vision & Community
Language-Agnostic Engine
Swap Dataset (e.g., ASL, BSL)
Worldwide Deployment
Community-Driven Model
Crowdsourcing, Open Standard (.ishara)
Goal: Building the scalable infrastructure to digitize sign language globally.

Conclusion

•**System Development:** Engineered a real-time, offline Android application translating ISL gestures into audible speech.
•**Core Technology:** Leveraged Google MediaPipe’s 21-point skeletal tracking for privacy-first, geometry-based recognition .
•**Data Innovation:** Created and validated a custom dataset for ISL alphabets and social gestures to bridge the resource gap.
•**Algorithmic Robustness:** Implemented "Wrist-Scaled Normalization" to ensure accuracy regardless of hand size or camera distance.
•**Performance Metrics:** Achieved 99.95% accuracy with sub-30ms latency, proving viability for standard mobile hardware .
•**User Experience:** Integrated Text-to-Speech (TTS) to successfully close the communication loop for deaf individuals.
•**Social Impact:** Delivered a scalable, zero-cost assistive solution accessible to anyone with a standard smartphone.

- **63 Million (India Hearing Impaired):**
  Source: World Health Organization (WHO) & Ministry of Health.
  Verification: who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss
  <300 Certified Interpreters:
  Source: Indian Sign Language Research & Training Centre (ISLRTC).
  Verification: islrtc.nic.in/certified-professionals
  430 Million (Global Need):
  Source: WHO World Report on Hearing (2021).
  Verification: who.int/publications/i/item/world-report-on-hearing

  Vision Architecture (MediaPipe):
  Paper: "MediaPipe: A Framework for Building Perception Pipelines" (Lugaresi et al., 2019).
  Source: arxiv.org/abs/1906.08172
  Sign Language Dataset Standards:
  Dictionary: ISLRTC Official ISL Dictionary (2021 Edition).
  Source: islrtc.nic.in/content/isl-dictionary